

Andres Salazar

Backend & Cloud Engineer · Python & GCP Serverless · Machine Learning in Production

Brighton, UK (open to relocating UK-wide) · +44 7300 562915 · andressalazar.alturo@gmail.com ·
linkedin.com/in/andres-salazar-alturo

Full UK work rights now (Graduate visa). New Entrant status: lower salary threshold and low cost to sponsor.

PROFESSIONAL SUMMARY

Backend and cloud engineer who builds and ships production systems end to end, from Python services and event-driven GCP serverless infrastructure to the machine-learning models that run on top of them. As the sole engineer at an early-stage startup, built the first working product single-handed, deployed it live to investors and stakeholders, and now runs it in real-world pilots. Strong in Python, FastAPI and Flask APIs, Google Cloud (Cloud Functions, Firebase) and the full delivery cycle: design, build, test, deploy and debug. MSc Artificial Intelligence and Adaptive Systems (Distinction), University of Sussex. Comfortable owning a whole system, learning fast, and delivering under pressure and to deadline.

TECHNICAL SKILLS

Languages: Python (primary), Java, C++, SQL.

Backend & APIs: FastAPI, Flask, REST APIs, event-driven and asynchronous services, integration and troubleshooting.

Cloud & serverless: Google Cloud Platform, Cloud Functions and triggers, Firebase (Auth, Firestore, Storage); IoT device-to-cloud integration.

Machine learning & data: data preprocessing, feature engineering, model training, validation and deployment; Pandas, NumPy, scikit-learn, TensorFlow.

DevOps & practices: Docker, Linux (Raspberry Pi), Git and GitHub, CI/CD, SDLC, code review, secure access patterns, testing and debugging.

PROFESSIONAL EXPERIENCE

AI & Cloud Engineer

Eluxevo Ltd, early-stage technology startup, Brighton, UK

Sep 2024 to Present

- Sole engineer for the product: designed, built and deployed the first working system end to end, and it remains the core the company has built every later feature on top of.
- Built and maintained the Python backend and event-driven GCP serverless infrastructure (Cloud Functions and triggers, Firebase Auth, Firestore and Storage) that connects multiple IoT devices to the cloud for remote control and local and cloud data capture.
- Deployed the local services in Python on a Linux-based Raspberry Pi and integrated them with the cloud, so devices run reliably both online and offline.
- Owned the machine-learning pipeline end to end: preprocessed the data, then trained, tested, validated and deployed the models that run in the live product.
- Deployed the live product demonstrations single-handed at Elemental London and BridgeAI, presenting to investors and industry stakeholders over two days and resolving issues on the spot, as the only person who knew the system end to end.
- Runs the current real-world pilots in people's homes, owning the full local-to-cloud data path and making the engineering decisions that keep them running.
- This work underpinned the company's visibility and traction and helped secure early grant funding, strengthening its position to raise a seed round.
- Designed, built, tested and deployed the initial Arduino and ESP-based electronic circuits for local device control and data capture during the first phase of the build.

- Works closely with the founder, front-end developers, designers and data analysts, coordinating progress and delivering consistently on time.

Lead Campus Coordinator

Kiwibot, autonomous delivery robotics, Ohio, USA

Aug 2022 to Dec 2022

- Led the campus rollout and daily operations, coordinating teams and resources to keep an autonomous delivery service running reliably.
- Supported fleet maintenance and worked with technical and data teams to diagnose and resolve operational issues.

Research Assistant

Universidad de Ibagué, Colombia

Jan 2020 to Mar 2021

- Built machine-learning datasets from multispectral, RGB and thermal imagery, engineered descriptors and validated regression models for crop-yield prediction.
- Ran iterative analysis and testing to improve model performance, and presented the work as a speaker (second place, university research week).

EDUCATION

MSc Artificial Intelligence & Adaptive Systems (Distinction)

University of Sussex, UK

2023 to 2024

Dissertation: neuroevolutionary swarm robotics with novelty search (Distinction).

BEng Electronic Engineering

Universidad de Ibagué, Colombia

2015 to 2021

ADDITIONAL

- Exhibitor at Elemental London (ExCeL) and BridgeAI with Eluxevo, and participant in the Evolve technology event (Silicon Brighton). Active in the Brighton software and AI meetup community.
- Languages: Spanish (native), English (fluent). Right to work: full UK work rights now on the Graduate visa; New Entrant for future sponsorship. Brazilian jiu-jitsu purple belt.